THE KERNEL OF A MAP OF ADEQUATE FUNCTORS IS ADEQUATE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The cokernel of a map of adequate functors is adequate, with the explicit kernel representation. Same chapter, Lemma lemma-cokernel-adequate, invoked at proof lines 725-728. Realises F as a kernel of a map of module-valued functors.

Source reference. Original lemma 06V2: sources/stacks-adequate/pinned-original-source.tex, source lines 684–684. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let $\varphi : F \to G$ be a map of adequate functors on Alg_A. Then* $\mathrm{Ker}(\varphi)$ *is adequate.*

Proof. Choose an injection $F \to \underline{M}$ and an injection $G \to \underline{N}$. Denote $F \to \underline{M \oplus N}$ the diagonal map so that

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \\ \underline{M \oplus N} & \longrightarrow & \underline{N} \end{array}$$

commutes. By Lemma 06V2 we can find a module map $M \oplus N \to K$ such that F is the kernel of $\underline{M \oplus N} \to \underline{K}$. Then $\mathrm{Ker}(\varphi)$ is the kernel of $\underline{M \oplus N} \to \underline{K \oplus N}$. □

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.